

The GRIM-pro family: two ways grammar-aware recombination gets through

Backdrop. *GRIM-pro* (GRIM-P) collects the roadblocks where *grimoire* — a structure-aware mutator that learns coherent input fragments (its `GeneralizationStage` gaps-and-preserves whole runs) and *splices/recurses* them back together — gets through, while a byte-level fuzzer stays stuck. Where I2S-pro is “the key was lying beside the lock” (paste *one* exact literal), GRIM-pro is “assemble the whole grammatically valid structure”: not one token, but a *recombination of several coherent constructs* into a larger, deeper input the parser/shaper will actually walk into. Both categories below share that move; they differ only in whether the advantage shows up as a *countable richer set of structural tokens* or as *sheer size/depth* with no countable tag vocabulary. Counts are validated roadblocks (n); the family total is 11.

Why the loser matters. For most of these branches the *blocking* arm is `cmplog` (the decisive pair is `grimoire` > `cmplog`), and that makes the test unusually sharp: `cmplog`’s input-to-state *can* paste an individual 4-byte table-tag literal directly. So if the gate were cleared merely by *placing a literal*, `cmplog` would match `grimoire` and the token advantage would vanish. A positive token lift *against the I2S arm* is what isolates grammar-aware *multi-token recombination* from plain literal substitution.

A note on the metric. All of GRIM-P is scored by one registry tool, `joint_necessity` in its generic two-arm `analyze_pair` mode, repointed to (winner = `grimoire`, loser = `cmplog` or `naive`). It reports `grimoire_tags` (mean number of *distinct* structural tokens per `grimoire` seed), `tag_lift` (that mean *minus* the loser’s), and `size_lift` (median seed-size *ratio* `grimoire`/loser). The token vocabulary is a per-target *measurement parameter*, not a new tool: for `harfbuzz` it is the 4-byte sfnt table tags (+ `ttcf`/`OTTO` magic), for `libxml2` the XML/DTD constructs (`<?xml`, `<!DOCTYPE`, `xmlns`, `<!ELEMENT`, `/>`, ...).

1. Grimoire builds more structure `grimoire.structural.assembly` ($n = 11$)

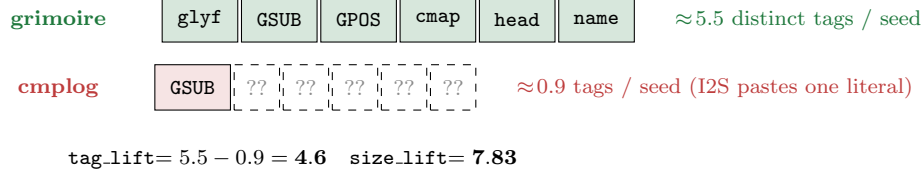
`grimoire`’s grammar-aware recombination clears the gate by assembling *more structure* than the byte/I2S loser. This surfaces through **two measurement routes**: a countable *token* lift (route A — richer distinct structural tokens, surviving even against the I2S arm) or, where there is no tag vocabulary to count, sheer recombined *size/depth* (route B). Same mechanism (grammar recombination assembles structure), two handles; which fires is a property of the gate. (We earlier split these as two categories — *structural_token_assembly* (9) and *structural_size_depth* (2) — but they are one mechanism with two routes, merged 2026-06-26.)

Route A — token assembly (`tag_lift`). The gate is cleared only once the input carries *several* coherent structural constructs at once — a font with multiple valid sfnt tables so the shaper resolves a real script, or a nested XML document with namespace-bearing attributes so the SAX path is actually walked. `grimoire` builds these by splicing learned fragments; the byte/I2S loser places few or none.

```
// harfbuzz hb-ot-shaper.hh : hb_ot_shaper_categorize()
// dispatch needs planner->props.script to resolve to a real complex
// script
switch (script) {
...
case HB_SCRIPT_GRANTHA: // <-- blocker (harfbuzz/6181): needs a
                        // structurally valid multi-table font
}
}
```

How we know it is recombination, not a pasted literal. cmplog (the blocking arm here) can splice a single 4-byte table tag via I2S. If that were enough, its seeds would carry as many distinct sfnt tokens as grimoire’s and tag_lift would collapse to 0. Instead the grimoire corpus carries *several more* distinct tokens per seed, and is much larger — the signature of assembling a coherent multi-table structure rather than dropping one literal.

Tool: joint_necessity (analyze_pair, winner=grimoire / loser=cmplog|naive)
Metric: grimoire_tags (mean distinct structural tokens per seed), tag_lift = grimoire_tags - loser_tags, size_lift = median size ratio grimoire/loser
Verify rule (the arbiter fires this, no LLM): tag_lift >= 1.0 AND grimoire_tags >= 2 AND size_lift >= 1.0 (cmplog route); tag_lift >= 1.0 AND naive_tags < grimoire_tags AND grimoire_tags >= 2 (naive route)
Real example: harfbuzz/6181 (hb_ot_shaper_categorize, case HB_SCRIPT_GRANTHA) — vs the I2S arm: grimoire_tags= 5.5 but cmplog_tags= 0.9, so tag_lift= 4.6 with size_lift= 7.83. The seven libxml2 branches (6924/6925/6926/6948/6950/7147/7327) take the *naive* route, e.g. 6924 (xmlNewNs, nested-namespace path): grimoire_tags= 6.1 vs naive_tags= 2, tag_lift= 4.1.



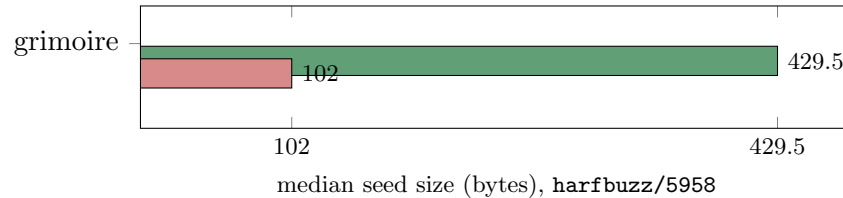
harfbuzz/6181: I2S can splice the *one* GSUB tag it logged, but it cannot *recombine* a coherent multi-table font. joint_necessity counts the distinct structural tokens each corpus carries; the gap between 5.5 and 0.9 is the grammar-recombination advantage that survives even against an I2S arm.

Route B — size / depth (size_lift). The fallback handle. The gate is about state-machine *depth* or repeated category structure rather than a countable target vocabulary, so the distinct-tag count barely moves — yet grimoire still wins because its recombined seeds are materially *larger and deeper*, driving the parser/shaper further into the input. We consult this rule only when the token-assembly test above does *not* fire, and it keys on size alone.

```
// harfbuzz hb-ot-shaper-thai.cc : get_consonant_type()
if (u == 0x0E0Eu || u == 0x0E0Fu) // <-- blocker (harfbuzz/5958):
                                // needs a 0x0E0E/0x0E0F run
                                // inside a
                                // deep generalized glyph-buffer
                                // scaffold
```

The gate needs a specific Thai consonant codepoint reached only deep inside a recombined glyph buffer. There is no table-tag vocabulary to count (tag_lift≈0); the discriminating signal is that grimoire’s seeds are several times larger than the loser’s.

Tool: `joint_necessity (analyze_pair, winner=grimoire / loser=cmplog)`
Metric: `size_lift` = median grimoire seed size / median loser seed size (consulted only when the token-assembly rule does not fire)
Verify rule (the arbiter fires this, no LLM): `size_lift >= 1.3`
Real example: `harfbuzz/5958 (get_consonant_type:57)` — `size_lift`= 4.21 (grimoire median 429.5 B vs cmplog 102 B) with `tag_lift` only 0.7. The other member, `libxml2/6743 (xmlParseStartTag2, attribute-defaulting path)`, has `size_lift`= 2.22 with *no* positive tag lift (−1.6): pure depth, not token richness.



When there is no tag vocabulary to count, sheer recombined size/depth ($4.21\times$ here) is what carries grimoire past the gate.

At a glance

Category	<i>n</i>	Example	One-line reason
structural assembly	11	harfbuzz/6181 (A), harfbuzz/5958 (B)	grammar recombination assembles more structure — route A recombine several coherent tokens, surviving even the I2S arm (<code>tag_lift</code> ≥ 1); route B sheer recombined size/depth when no tag vocabulary (<code>size_lift</code> ≥ 1.3)
GRIM-P total	11		all scored by <code>joint_necessity analyze_pair</code> , winner = grimoire

All counts and metric values are from `bench/dataset.jsonl` and `step5b_new_v3/grimoire_structural_WL/assignments_{s4,sB}.json`; the verify rules are the deterministic arbiter rules (no LLM in the decision). The harfbuzz corpora live on server `s4`, the libxml2 corpora on server `sB`; gates are quoted from the per-branch analysis cards.